EXPERIMENT NO 13

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13.1.1 Fibonacci sequence using modules

Algorithm

Step 1: Start

Step 2: Input integer n

Step 3: Check if n >= 0

- If **No**, print *"Please enter a non-negative integer"* and go to Step 10
- If **Yes**, go to Step 4

Step 4: Call function generate_fibonacci_sequence(n)

Step 5: Inside function, check if n < 0

- If **Yes**, return empty list

Step 6: Initialize seq = [], a = 0, b = 1

Step 7: Repeat while a <= n

- Add a to seq
- Update values: a, b = b, a + b

Step 8: Return seq to main program

Step 9: Print all elements of returned Fibonacci sequence

Step 10: End

Flowchart

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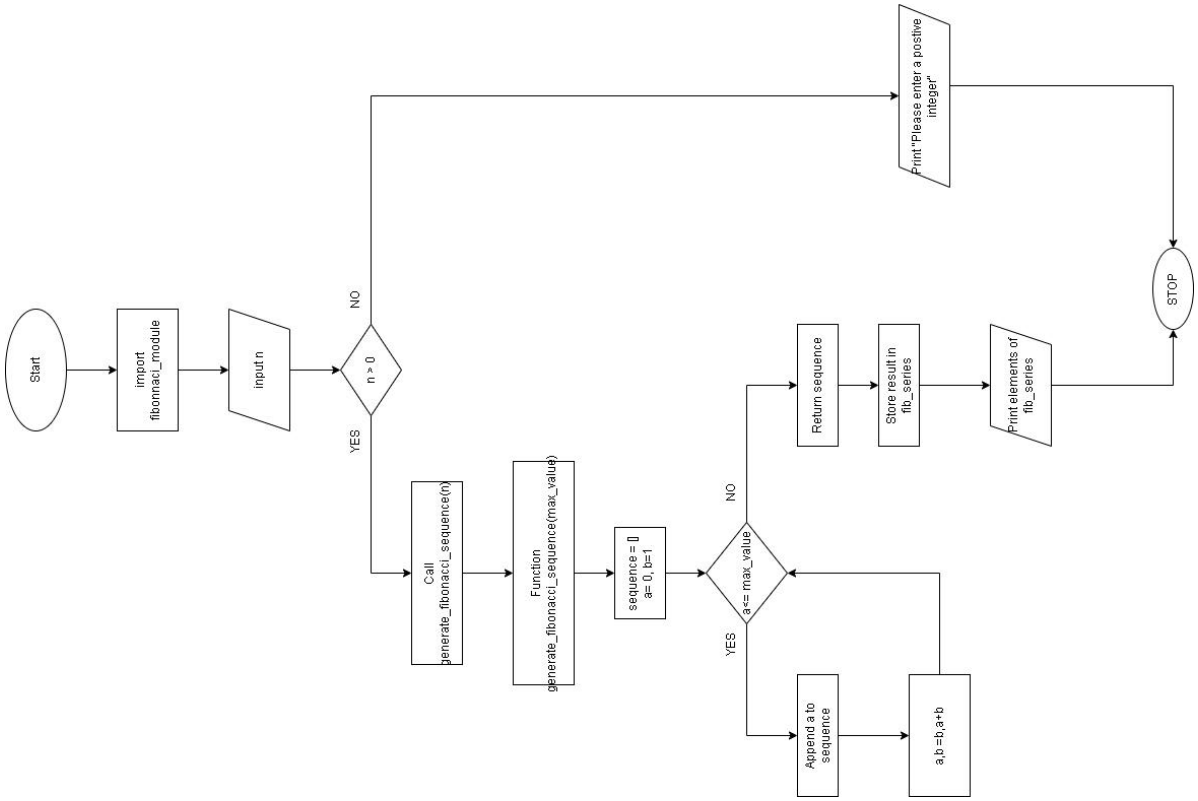

Python code

```
# Write your code here.....
def generate_fibonacci_sequence(max_value):
    if max_value < 0:
        return []
    seq = []
    a=0
    b=1
    while a<=max_value:
        seq.append(a)
        a,b=b,a+b
    return seq
# Write your code here......
import fibonacci_module
n = int(input())
```

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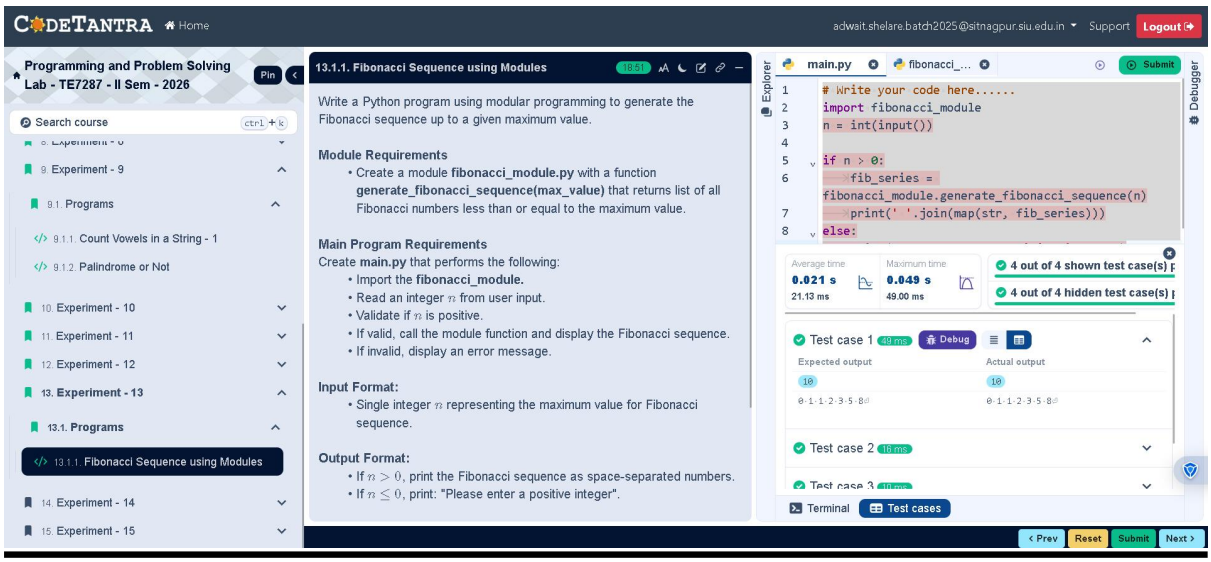
```
if n > 0:
        fib_series = fibonacci_module.generate_fibonacci_sequence(n)
        print(' '.join(map(str, fib_series)))
else:
        print("Please enter a positive integer")
```

EXCECUTION